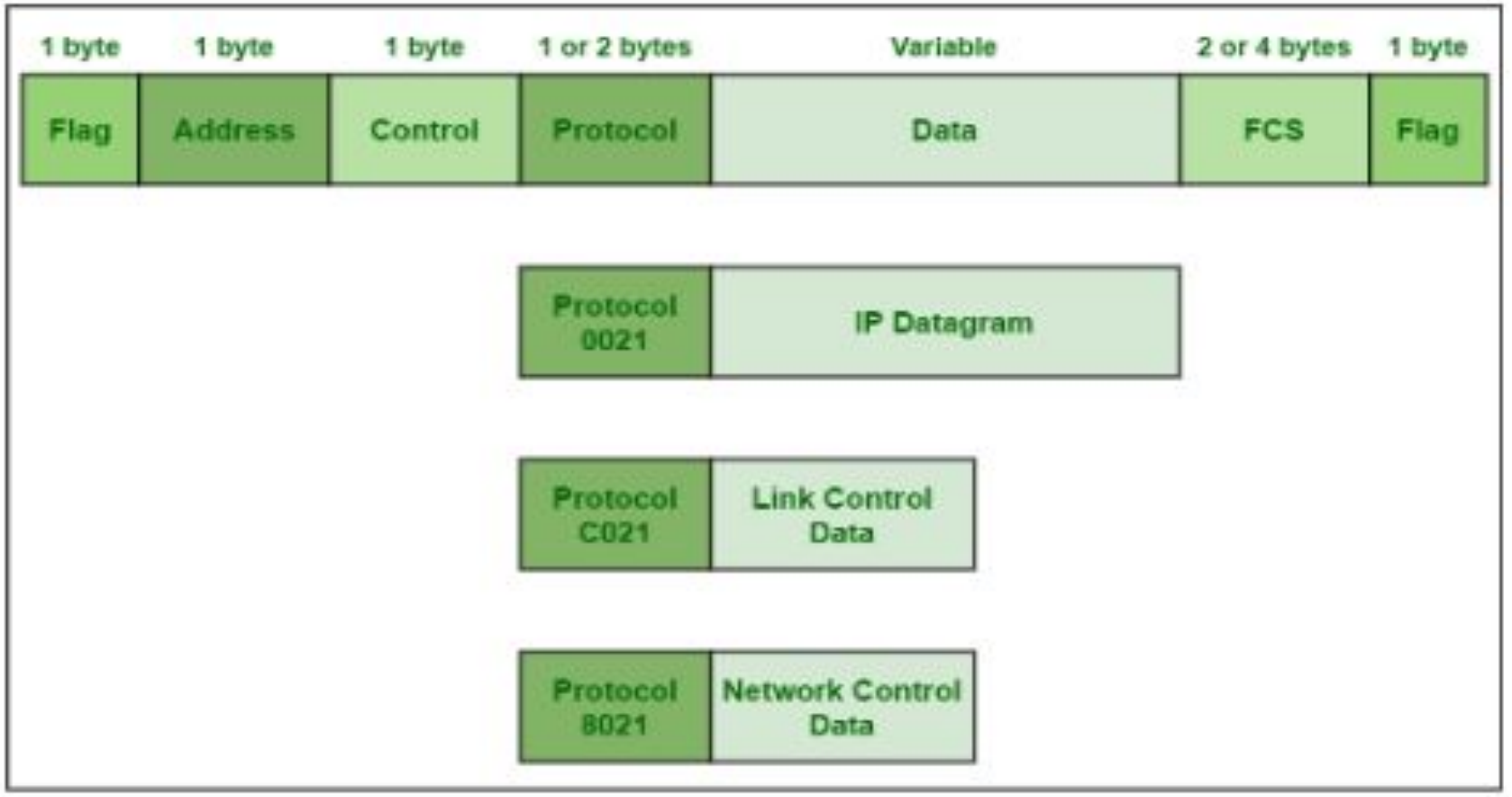
Point to Point Protocol

- PPP is generally the default RAS protocol in windows and is most commonly used protocol of data link layer that is required to encapsulate higher network-layer protocols simply to pass over synchronous and asynchronous communication lines.
- In PPP, link establishment is controlled and handled mainly by Link Control Protocol (LCP). It is also required to connect the Home PC to server of ISP through a modem. It was also adopted by ISPs to simply provide dial-up Internet Access.
- **PPP Frame Format :** PPP frame is generally required to encapsulate packets of information or data that simply includes either configuration information or data. PPP basically uses the same basic format as that of HDLC.
- PPP usually contains one additional field i.e. protocol field. This protocol field is present just after control field and before information or data field.

PPP Frame Format



PPP Frame Format

Various fields of Frame are given below :

- **Flag field -** PPP frame similar to HDLC frame, always begins and ends with standard HDLC flag. It always has a value of 1 byte i.e., 01111110 binary value.
- **Address field -** Address field is basically broadcast address. In this, all 1's simply indicates that all of the stations are ready to accept frame. It has the value of 1 byte i.e., 11111111 binary value. PPP on the other hand, does not provide or assign individual station addresses.
- **Control field -** This field basically uses format of U-frame i.e., Unnumbered frame in HDLC. In HDLC, control field is required for various purposes but in PPP, this field is set to 1 byte i.e., 00000011 binary value. This 1 byte is used for a connection-less data link.
- **Protocol field -** This field basically identifies network protocol of the datagram. It usually identifies the kind of packet in the data field i.e., what exactly is being carried in data field. This field is of 1 or 2 bytes and helps in identifies the PDU (Protocol Data Unit) that is being encapsulated by PPP frame.
- **Data field -** It usually contains the upper layer datagram. Network layer datagram is particularly encapsulated in this field for regular PPP data frames. Length of this field is not constant rather it varies.
- **FCS field -** This field usually contains checksum simply for identification of errors. It can be either 16 bits or 32 bits in size. It is also calculated over address, control, protocol, and even information fields. Characters are added to frame for control and handling of errors.

HDLC and PPP Comparison

Feature	PPP	HDLC
Full Form	Point-to-Point Protocol	High-Level Data Link Control
Standardization	Open standard (RFCs)	ISO standard (vendor-dependent in practice)
Communication Type	Point-to-point only	Point-to-point & multipoint
Authentication	✔ Supports PAP & CHAP	❌ No authentication
Error Detection	✔ CRC	✔ CRC
Error Correction	❌ Not provided	❌ Not provided
Flow Control	❌ No	✔ Yes
Network Layer Support	✔ Multiple protocols (IP, IPv6, etc.)	❌ Mainly IP
Configuration	✔ Automatic (LCP, NCP)	❌ Manual
Vendor Dependency	❌ Vendor-independent	✔ Vendor-specific implementations
Overhead	Higher	Lower

Media Access Control (MAC)

- Media Access Control (MAC) is a sublayer of the Data Link Layer (Layer 2) in the OSI model that manages how devices on a network gain access to a shared medium to transmit data.
- It prevents data collisions, ensures orderly communication in wired (Ethernet) and wireless (Wi-Fi) networks, and uses unique hardware addresses to identify devices.

Key Aspects of Media Access Control:

- **Purpose:** Coordinates access to a shared transmission medium, acting like a chairperson managing speakers in a meeting to prevent simultaneous transmission.
- **Sublayer of Data Link Layer**: MAC resides between the physical layer and the Logical Link Control (LLC) sublayer.
- **Collision Prevention:** Protocols like CSMA/CD (for wired) and CSMA/CA (for wireless) are used to prevent data collisions, which cause communication breakdowns.
- **Addressing:** Each network interface controller (NIC) is assigned a unique 48-bit or 64-bit MAC address to ensure data reaches the correct destination within a network segment.
- **Functions:** Responsibilities include framing (identifying start/end of data), managing channel access, and error detection.

Main Methods/Types:

- **Contention-based:** Nodes compete for access (e.g., Ethernet using CSMA/CD).
- **Controlled-access:** Nodes take turns (e.g., token passing).
- **Channelization:** Bandwidth is divided (e.g., FDMA, TDMA, CDMA).